SWE 307 BIG DATA PROJECT - 2

Redis for RDB caching, HDFS as Data Storage and Local LLM (Ollama) Server for chatting

Due date: 5.11.2025 Thursday, in class.

This project aims to build a real-time information system that combines the speed of in-memory data processing with the reliability and scalability of distributed storage (See Figure 1.). The system uses **Redis** to handle fast, real-time ingestion and temporary caching of incoming data, while **Hadoop HDFS** provides a distributed and fault-tolerant file system to store documents that are uploaded or downloaded. The Hadoop HDFS is expected to improve response time as well. This semester an LLM server (Ollama 3.2) is added to the system just for chatting from front-end.

Key Objectives:

- Use scott database (https://github.com/rsp/pg-scott) for RDBMS(MySQL). At the beginning, import data from given site to your database.
- Use **Redis** to temporarily store and process incoming data for fast access.Periodically flush the data from Redis to **RDBMS** for durable storage and future processing. Show that the caching mechanism works as expected. (See example: https://www.youtube.com/watch?v=0a-RlJx09rg)
- Use **Hadoop-HDFS** for document storage.
- Use **Ollama 3.2** as local LLM server. One model (llama3.2:latest) would be sufficient.

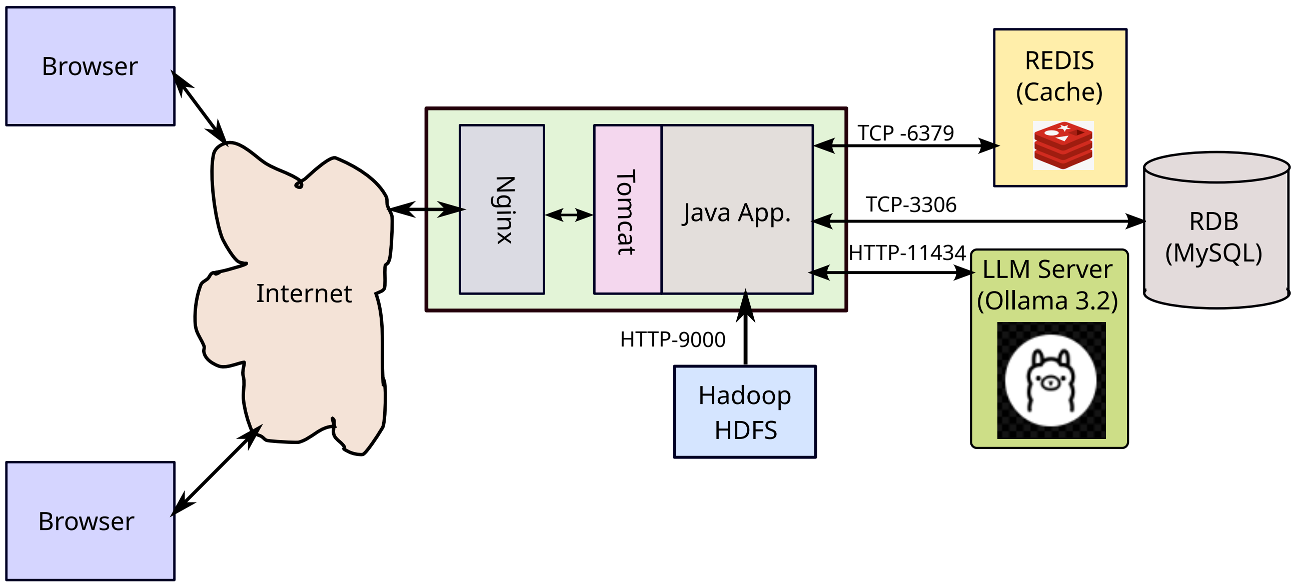

Figure 1. Architectural block diagram of project 2.

What is required from you is as follows:

1) Single node Hadoop-HDFS cluster must be installed in your computer.

2) Redis and MySQL must be installed on your computer.

3) Ollama 3.2 server must be installed on your computer.

4) A simple Java Spring-Boot application will be developed to perform the following tasks:

a) Personnel and department data will be read/updated from the database via Redis.
b) Personnel images will be stored to/read from HDFS.
c) There will be two pages in this application: 1) Personnel information will be displayed in a table using the JOIN operation on database. Information to display: employee image, employee name, manager name, salary, commission, department name.

d) On the second page (chat page), we should be able to chat with LLM server.

Notes:

1) Some groups may want to implement the project using G-Drive or AWS-S3 Object storage instead of HDFS, these implementations will also be accepted. This will be discussed in the class.
2) Example image files and data files will be provided on Github repository, you can clone/download everything provided.
Link: https://github.com/ozmen54/SWE307-2026.git

Details will be discussed in the class!